



GSV2001

1 In to 2 Out HDMI2.0 Repeater with Audio
Extraction/Insertion

September, 2019

PRODUCT SPECIFICATION

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Glossary

Term	Definition
DDC	Display Data Channel
EDID	Extended Display Identification Data
ESD	Electrostatic Discharge
HDCP	High-bandwidth Digital Content Protection
HDMI	High-Definition Multimedia Interface
HPD	Hot Plug Detect
I ² C	Inter-Integrated Circuit
I ² S	Integrated Interchip Sound
MCU	Microcontroller Unit
MISO	Master In Slave Out
MOSI	Master Out Slave In
OTP	One Time Programmable
PCM	Pulse Code Modulation
S/PDIF	Sony/Philips Digital Interface Format
SPI	Serial Peripheral Interface
TMDS	Transition Minimized Differential Signaling
SCDC	Status and Control Data Channel
CEC	Consumer Electronics Control
CSC	Color Space Conversion

1 General Information

1.1 General Information

The GSV2001 is HDMI1.4/2.0 compatible, HDCP 1.4/2.2 supported, configurable 1-in-2-out repeater device. All 2 outputs are identical on transmitter capability.

The HDMI input and output maximum processing pixel clock frequency is 600MHz which means the video resolution can support up to 4kx2k@60Hz 4:4:4 8-bit. The maximum processing audio sample frequency is 192K Hz for non-compression timing. GSV2001 supports HDR10 and Dolby Vision HDR as input and output.

For audio insertion and extraction, the versatile TTL pin bus of GSV2001 can be configured as either input mode or output mode regarding platform requirement. GSV2001 can support up to 8-channel I2S or 2-channel S/PDIF, 3D and multi-stream audio. In TDM mode, each audio pin supports up to 8 channels.

Internal Scaler and Color Space Converter enables the input and output to be timing format independent and capable of long distance transmission.

With powerful HDMI Rx equalizer and Tx pre-emphasis capability, GSV2001 can cascade itself (or GSV2000 series chips) with at least 7 stages for all HDMI 1.3/1.4/2.0 timings.

An internal Clock Generator can be used to generate 4 independent output clocks, which will greatly remove complexity of using dedicated clock device.

1.2 Features

1.2.1 HDMI Video Input and Output

- Compliant with HDMI2.0b, HDMI1.4b
- Compliant with HDCP2.2/2.3 and HDCP1.4
- Data rate up to 18Gbps
- Programmable HDMI Tx output swing, slew-rate, pre-emphasis
- Adaptive receiver equalization
- AC-coupling capable
- Color Space Converter supports any conversion between different color spaces
- HDR supported (HDR10/HDR12/Dolby Vision/HLG)
- 5V tolerance on DDC/HPD/CEC
- Arbitrary video stream matrix between HDMI Rx and HDMI Tx

1.2.2 Audio Input/Output

- SPDIF/I2S/HBR/DSD/TDM Audio Extraction
- SPDIF/I2S/HBR/DSD/TDM Audio Insertion
- Configurable direction for each Audio bus
- Arbitrary audio stream matrix between HDMI Rx/HDMI Tx/Audio bus

1.2.3 Internal Downscaler

Scaler is only used to downscale 4k UHD timings to 2k FHD timings. The horizontal resolution and vertical resolution are both cut in half while frame rate remains the same.

1.2.4 Color Space Converter

Color Space Converter can convert RGB and YCbCr by the following table. It should be noted that YCbCr 422 shares the same color space with YCbCr 444 in internal routing. So any conversion that YCbCr 444 supports, YCbCr 422 also supports it.

Table 1. Color Space Converter Support Table

From	To	To
RGB	YCbCr 444	YCbCr 420
YCbCr 444	YCbCr 420	RGB
YCbCr420	YCbCr444	RGB

1.2.5 Clock Generator

Internal Clock Generator has 4 independent clock outputs ranging 25MHz~600MHz. Clock Generator generates output clocks based on divider/multiplier ratio of GSV2001's 27MHz oscillator. The no-compensation PLL circuit architecture minimizes output clock jitter. With fast PLL acquisition time, clock generator output clock frequency matches its ratio register settings in nano-seconds.

1.2.6 CEC

There is an internal CEC engine for handling CEC low level transactions. Customer design only needs to manipulate I2C registers for reading and writing CEC commands.

1.3 Chip Application Modes

1.3.1 Audio Extraction and Video Distribution

GSV2001 has 1 HDMI input port and 2 HDMI output ports. All of the ports are HDCP 1.4 and HDCP 2.2 capable.

One important purpose of extracting HDMI encrypted stream is to extract audio, process it in audio signal processing chip (DSP/FPGA e.g.). As shown below, TTL audio pins can be configured as output to implement this feature. It should be noted that extracted audio stream is routed from the selected HDMI input to distribute to HDMI outputs.

Limited by the only audio TTL bus, when audio extraction is implemented, audio insertion can not be implemented. So when using audio extraction, both Tx's audio can only be streamed from HDMI Rx rather than from external audio input.

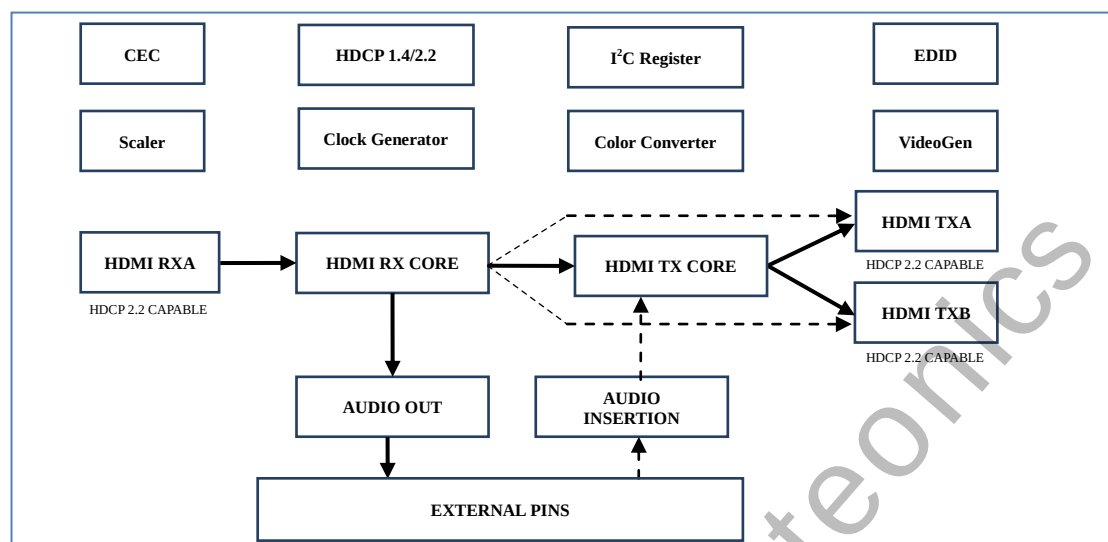


Figure 1. GSV2001 audio extraction diagram

1.3.2 Audio Insertion and Video Distribution

Inserted audio can be routed to any individual Tx or both Tx streams while HDMI video is still routed from HDMI Rx.

Limited by the only audio bus, when audio insertion is implemented, audio extraction can not be implemented. In this mode, both Tx's audio can be streamed either from HDMI Rx or external audio input.

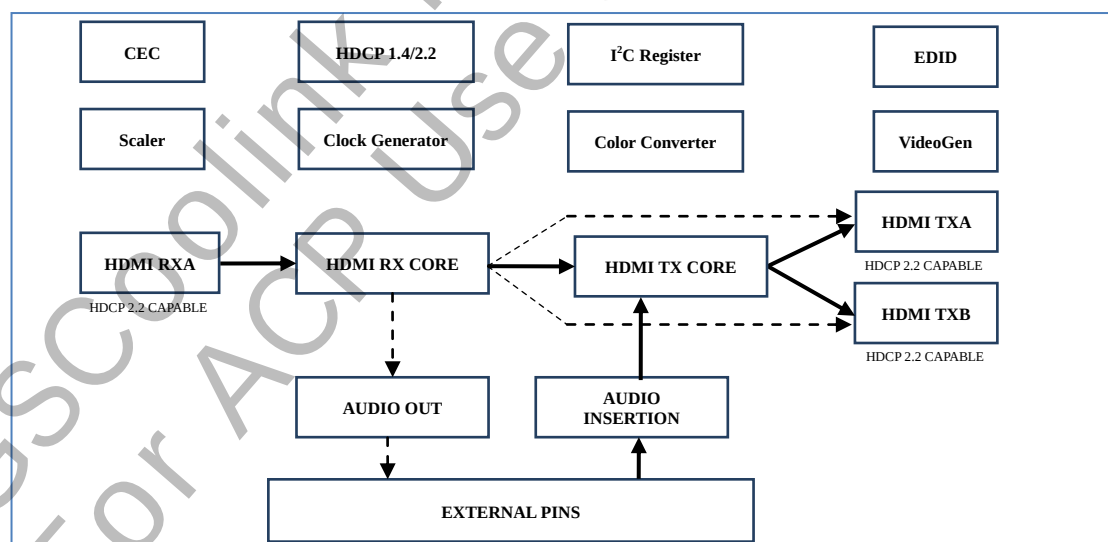


Figure 2. GSV2001 audio insertion diagram

1.3.3 HDMI 2.0 to HDMI 1.4 downscaler

GSV2001 has a built-in scaler and color converter. These blocks enable the chip to do internal video processing for matching HDMI source and sink's capability with a wider range and better performance. Flexible connection between IPs creates versatile

usage modes.

Here is an example of 4K YCbCr 420 (300MHz pixel clock) input, 4K YCbCr 420 (300MHz pixel clock) output and 1080p YCbCr 422 (150MHz pixel clock) output. It should be noted that Tx Core can still do color space conversion (except for YCbCr420) without Color Conversion block. This makes HDMI output flexible in color space.

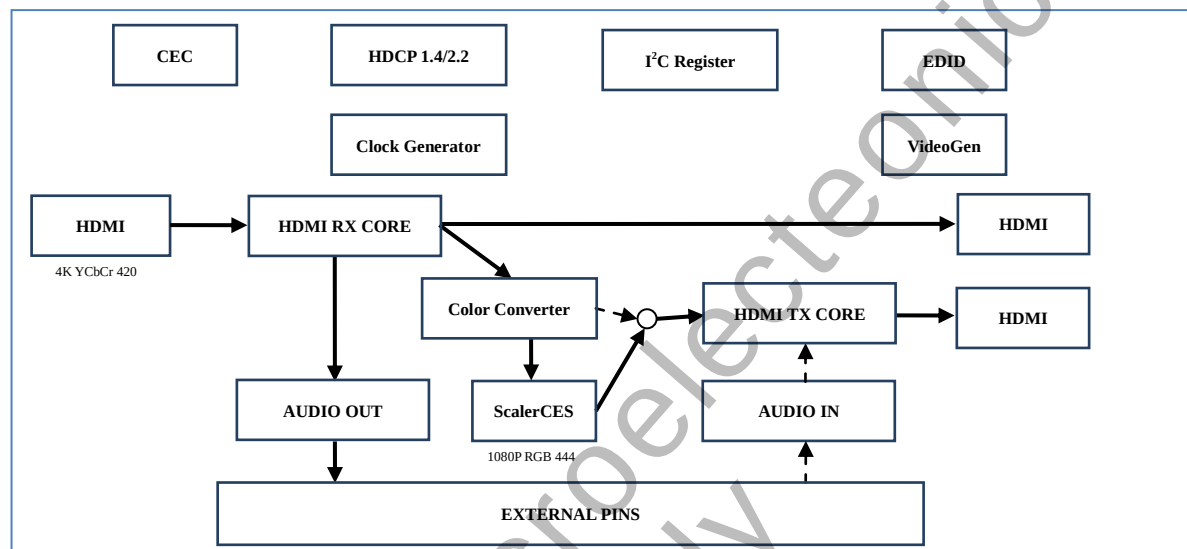


Figure 3. GSV2001 4K 420-to-1080p 422 diagram

1.3.4 HDMI 2.0 YCbCr 420 color space conversion

Here is an example of 4K RGB 444 (600MHz pixel clock) input, 4K YCbCr 422 (600MHz pixel clock) output and 4K YCbCr 420 (300MHz pixel clock) output. It should be noted that Tx Core can still do color space conversion (except for YCbCr420) without Color Conversion block. This makes HDMI output flexible in color space.

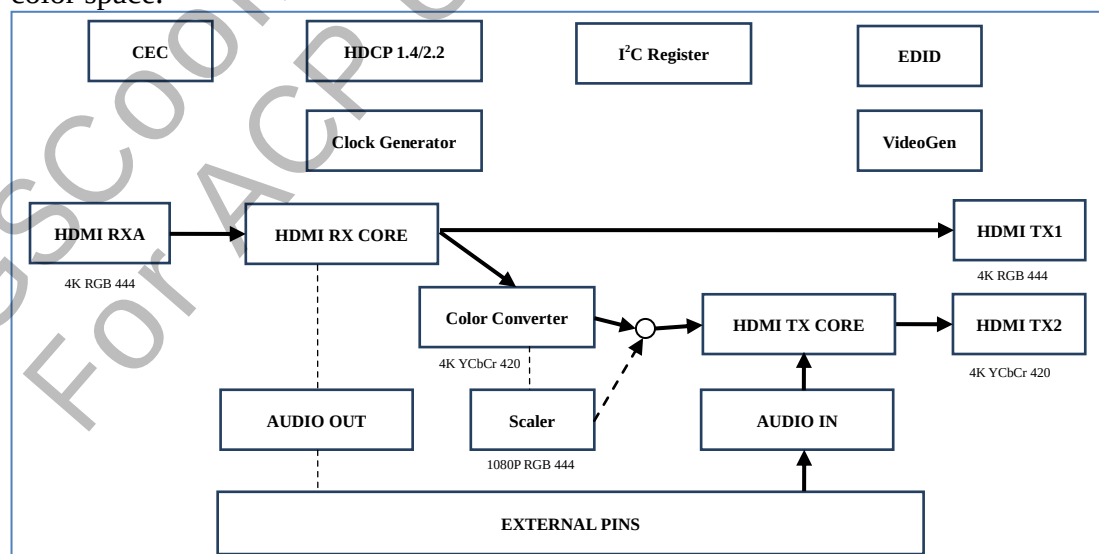


Figure 4. GSV2001 4K 444-to-4K 420 diagram

1.3.5 Enabling Clock Generator

When using Clock Generator to generate clock outputs, TxB's 4 differential pairs are occupied. The pins are internally pulled up to 3.3V with maximum 20mA differential current. Each PLL output clock has its independent ratio register setting.

4 pairs of GSV2001 differential output clock are recommended to be AC-coupled when connecting to device's differential clock input. If DC-coupled, GSV2001's differential output clock is internally pulled-up to 3.3V. In single-end clock application, each 50% duty-cycle differential clock output pair can be separated to clocks with 180 degrees phase shift.

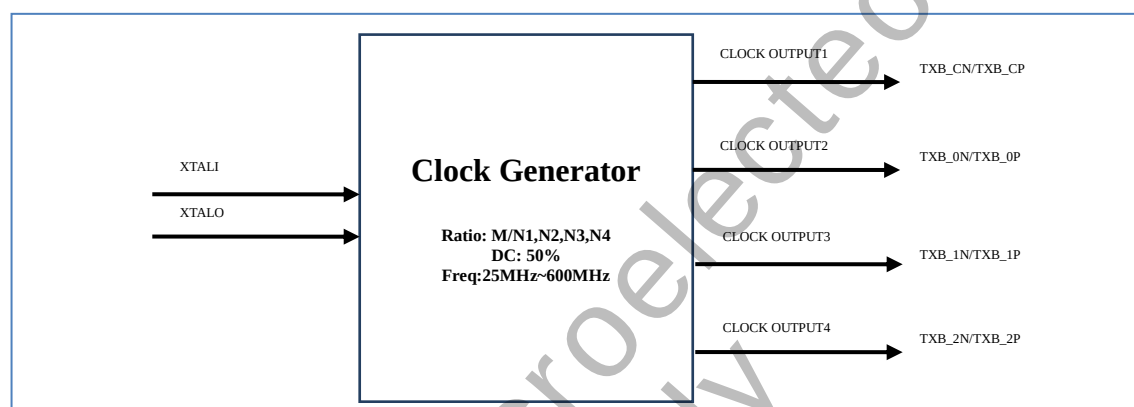


Figure 5. GSV2001 Clock Generator diagram

1.4 Audio Bus Output Capability

When one group of audio bus is configured as output, I2S and SPDIF are output at the same time. Normal Pin Setting is as below.

Table 2. I2S/SPDIF Audio Extraction

Pin Name	Alias	Direction	Description
AP0	SDATA[0]	Output	I2S Data, default stereo channels
AP1	SDATA[1]	Output	I2S Data, 3/4 channels
AP2	SDATA[2]	Output	I2S Data, 5/6 channels
AP3	SDATA[3]	Output	I2S Data, 7/8 channels
AP4	SPDIF	Output	SPDIF channel
AP5	LRCLK/WS	Output	Fs (0 = Left, 1 = Right)
SCLK	BCLK	Output	Fixed to 64Fs
MCLK	Sys Clock	Output	Selected from 128Fs/256Fs/384Fs/512Fs

For HBR application, SPDIF are also capable of sending out with 4 pins.

Table 3. HBR Audio Extraction in SPDIF

Pin Name	Alias	Direction	Description
AP0	SPDIF[0]	Output	SPDIF Data[0], 1/2 channels
AP1	SPDIF[1]	Output	SPDIF Data[1], 3/4 channels
AP2	SPDIF[2]	Output	SPDIF Data[2], 5/6 channels

AP3	SPDIF[3]	Output	SPDIF Data[3], 7/8 channels
AP4	SPDIF	Output	SPDIF channel, 1/2 channels

1.5 Audio Bus Input Capability

When Audio Bus is set to Input, either I2S or SPDIF can be selected as audio input source. It should be noted that MCLK must be externally fed into GSV chip in I2S audio insertion mode.

For SPDIF input, GSV series chip can detect Sampling Frequency and automatically update it in Channel Status with GSV software. For I2S input, software designer needs to indicate the current input sampling frequency in software. Common application modes are listed.

Table 4. Stereo I2S Input

Pin Name	Alias	Direction	Description
AP0	SDATA[0]	Input	I2S Data, default stereo channels
AP5	LRCLK/WS	Input	Fs (0 = Left, 1 = Right)
SCLK	BCLK	Input	Fixed to 64Fs
MCLK	Sys Clock	Input	Selected from 128Fs/256Fs/384Fs/512Fs

Table 5. SPDIF Input

Pin Name	Alias	Direction	Description
AP0	SPDIF	Input	SPDIF channel

Table 6. 8 channel I2S Input

Pin Name	Alias	Direction	Description
AP0	SDATA[0]	Input	I2S Data, default stereo channels
AP1	SDATA[1]	Input	I2S Data, 2/3 channels
AP2	SDATA[2]	Input	I2S Data, 4/5 channels
AP3	SDATA[3]	Input	I2S Data, 6/7 channels
AP5	LRCLK/WS	Input	Fs (0 = Left, 1 = Right)
SCLK	BCLK	Input	Fixed to 64Fs
MCLK	Sys Clock	Input	Selected from 128Fs/256Fs/384Fs/512Fs

14	TXB_2N	O	TXB Negative TMDS differential line driver data output [2]
15	TXB_2P	O	TXB Positive TMDS differential line driver data output [2]
16	TXB_AVDD	Power	Analog 1.2V voltage power supply for TXB Port
17	TXB_HPD	I	TXB 5V tolerance HPD PAD
18	TXB_SDA	I/O	TXB 5V tolerance DDC SDA PAD
19	TXB_SCL	I/O	TXB 5V tolerance DDC SCL PAD
20	TXB_UTILILY	I	TXB 5V tolerance UTILITY PAD/TXB Audio Return Channel True Input
21	TXA_SDA	I/O	TXA 5V tolerance DDC SDA PAD
22	TXA_SCL	I/O	TXA 5V tolerance DDC SCL PAD
23	TXA_HPD	I	TXA 5V tolerance HPD PAD
24	TX_PVDD	Power	PLL 1.2V voltage power supply for TX
25	TXA_AVDD	Power	Analog 1.2V voltage power supply for TXA Port
26	TVDD	Power	Analog 3.3V voltage power supply
27	TXA_CN	I/O	TXA Negative TMDS clock output
28	TXA_CP	I/O	TXA Positive TMDS clock output
29	TXA_0N	O	TXA Negative TMDS differential line driver data output [0]
30	TXA_0P	O	TXA Positive TMDS differential line driver data output [0]
31	TXA_AVDD	Power	Analog 1.2V voltage power supply for TXA Port
32	TXA_1N	O	TXA Negative TMDS differential line driver data output [1]
33	TXA_1P	O	TXA Positive TMDS differential line driver data output [1]
34	TXA_2N	O	TXA Negative TMDS differential line driver data output [2]
35	TXA_2P	O	TXA Positive TMDS differential line driver data output [2]
36	TXA_AVDD	Power	Analog 1.2V voltage power supply for TXA Port
37	DVDD	Power	Digital 1.2V voltage power supply
38	DVDDIO	Power	Digital IO 3.3V voltage power supply
39	TXA_AVDD	Power	Analog 1.2V voltage power supply for TXA Port
40	TXA_AVDD	Power	Analog 1.2V voltage power supply for TXA Port
41	DVDDIO	Power	Digital IO 3.3V voltage power supply
42	TXA_AVDD	Power	Analog 1.2V voltage power supply for TXA Port
43	DVDD	Power	Digital 1.2V voltage power supply
44	AUD_D0	I/O	Digital IO PAD for Audio Data0
45	AUD_D1	I/O	Digital IO PAD for Audio Data1
46	AUD_D2	I/O	Digital IO PAD for Audio Data2
47	AUD_D3	I/O	Digital IO PAD for Audio Data3
48	AUD_D4	I/O	Digital IO PAD for Audio Data4
49	AUD_D5	I/O	Digital IO PAD for Audio Data5
50	AUD_SCLK	I/O	Digital IO PAD for Audio SCLK
51	AUD_MCLK	I/O	Digital IO PAD for Audio MCLK
52	I2C_SDA	I/O	Digital IO PAD for I2C Data
53	I2C_SCL	I/O	Digital IO PAD for I2C Clock
54	CSB	I/O	Chip Select Pin
55	RESETB	I	Reset Pin
56	INT	O	Interrupt Output Pin
57	AVMUTE	Q	AV-mute Output Pin
58	DVDD	Power	Digital 1.2V voltage power supply
59	RX_AVDD	Power	Analog 1.2V voltage power supply for RX port
60	TVDD	Power	Analog 3.3V voltage power supply
61	RXA_CN	I	RXA Negative TMDS clock input
62	RXA_CP	I	RXA Positive TMDS clock input
63	RXA_0N	I	RXA Negative TMDS differential data input [0]
64	RXA_0P	I	RXA Negative TMDS differential data input [0]
65	RX_AVDD	Power	Analog 1.2V voltage power supply for RX port
66	RXA_1N	I	RXA Negative TMDS differential data input [1]
67	RXA_1P	I	RXA Negative TMDS differential data input [1]
68	RXA_2N	I	RXA Negative TMDS differential data input [2]
69	RXA_2P	I	RXA Negative TMDS differential data input [2]
70	RX_AVDD	Power	Analog 1.2V voltage power supply for RX port
71	REXT400	I/O	External 400ohm resistor to TVDD
72	CEC	I/O	5V tolerance CEC PAD
73	RX_HPD	I/O	RX 5V tolerance HPD PAD
74	RX_SDA	I/O	RX 5V tolerance DDC SDA PAD

75	RX_SCL	I	RX 5V tolerance DDC SCL PAD
76	RX_5V	Power	RX 5V POWER

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3 Electrical Specifications

3.1 TMDS Transmitter

3.1.1 Transmitter DC and AC Specification

Table 8. TMDS DC and AC Characteristics

Item	Value
Single-Ended High Level Voltage Range: Data Channels 0,1,2	AVcc-400mV to AVcc+10mV
Single-Ended Low Level Voltage Range: Data Channels 0,1,2	AVcc-1000mV to AVcc-400mV
Single-Ended High Level Range: Clock Channel	AVcc-400mV to AVcc+10mV
Single-Ended Low Level Voltage Range: Clock Channel	AVcc-1000mV to AVcc-200mV
Single-Ended Swing Voltage: Data Channels 0,1,2	400mV ≤ Vswing ≤ 600mV
Single-Ended Swing Voltage: Clock Channel	200mV ≤ Vswing ≤ 600mV

3.1.2 Transmitter Characteristic

3.1.2.1 TMDS Output Current

- Range: 3mA~18mA, 1mA per step;
- Can be increased by 25% and 50%

3.2 Timing Information

3.2.1 Power Up and Reset Timing Diagrams

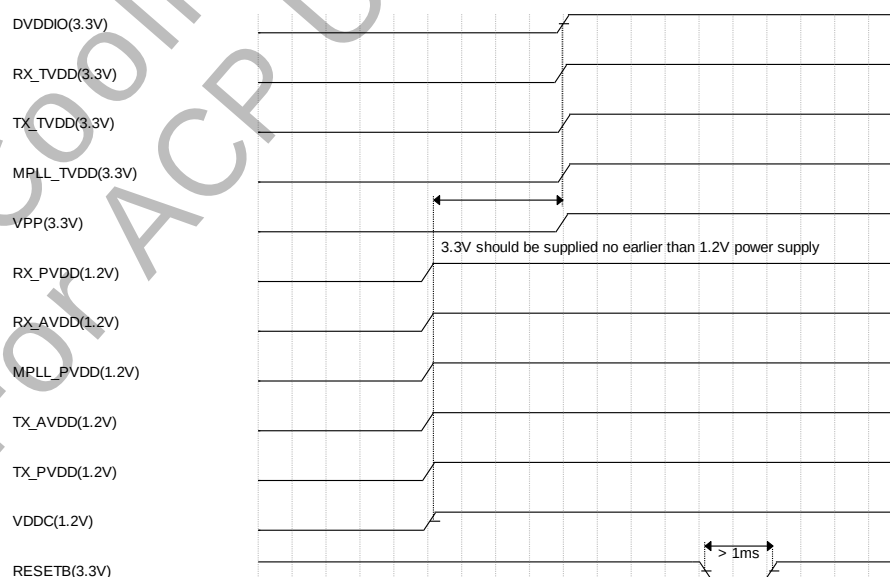


Figure 7. Power Up Sequence

3.2.2 I2C Timing Diagrams

The I2C bus used 8-bit page address and 16-bit register address. For every register, 8-bit data will be accessed.

The I2C write timing is shown below.

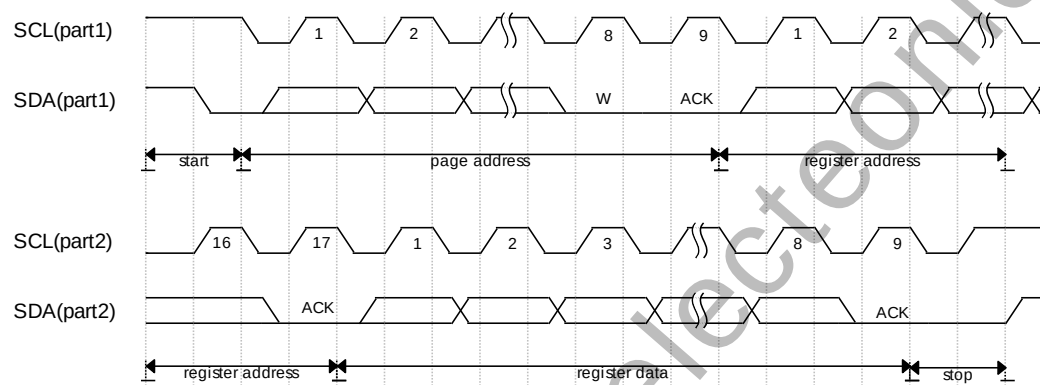


Figure 8. I2C Timing Diagram(Write)

The I2C read timing is shown below.

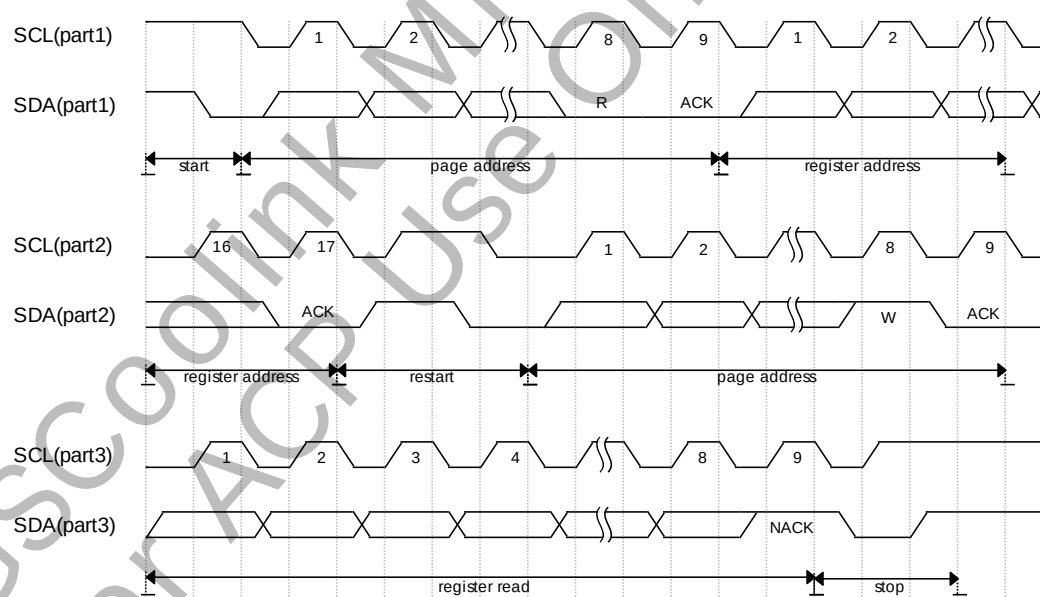


Figure 9. I2C Timing Diagram(Read)

3.3 Operating Conditions

3.3.1 Temperature Conditions

GSV2001's operation temperature range is -20 °C to 85 °C.

GSV2001's junction temperature can be calculated based on $T_j = T_c + \theta_{Jc} * P_D$.

Use T_c and θ_{Jc} , P_D of GSV2001 max power is 1.35W. θ_{Jc} is 4.4C/W. T_j can thus be calculated.

3.3.2 Power Conditions

GSV2001's power consumption is 1350mW for maximum performance configuration. Typical Case of 1-input, 2-outputs, 4K60 444 with HDCP 2.2 power consumption.

Table 9. 4K60 Power Consumption

GSV2001	Power Domain	Current Consumption	Power Consumption
3.3V	TVDD	150mA	0.495W
	DVDDIO	<1mA	
1.2V	DVDD	305mA	0.366W
1.2V	TX_AVDD	70mA	0.462W
	RX_AVDD	215mA	
	TX_PVDD	80mA	
	MPLL_PVDD	20mA	

3.3.3 Audio Pin Conditions

GSV2001's Audio TTL pins can tolerate 2.8V~3.6V as logic HIGH.

3.3.4 I2C Conditions

GSV2001's I2C maximum SCL frequency is 400KHz.

3.3.5 ESD Conditions

With RCLAMP0524 as external ESD, Rx TMDS and Tx TMDS can both pass 15kV.

4 Package Information

The GSV2001 device is packaged in a 76-pin, 9mmx9mm QFN76L package. There is an ePad as the electrical ground of the device. It is critical to solder the entire ePad firmly onto the PCB. A weak connection of the ePad could result in poor performance on higher frequency HDMI video timing.

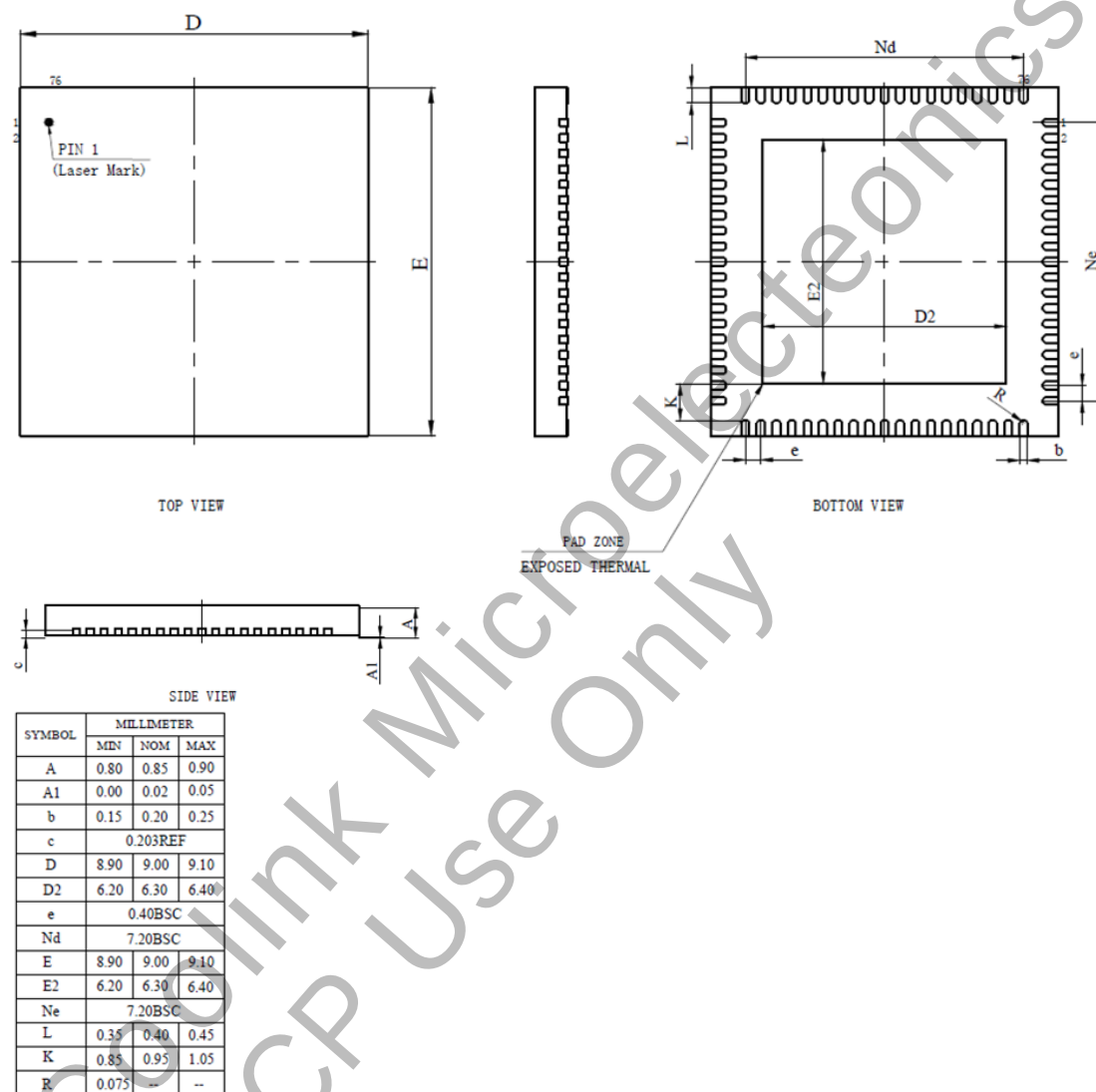


Figure 10. GSV2001 package dimensions

5 Ordering Guide

Table 10. Ordering Information

Part Number.	Temperature Range	Package Description	Packing Type
GSV2001	−20 °C to +85 °C	QFN76L, 0.4 mm ball pitch, 9 mm x 9 mm outline	Tray

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6 Revision History

Table 11. Revision history

Revision No.	Description	Date
V0.1	Draft Initial Version for internal review.	Aug 13, 2018
V0.2	Characteristics values updated, all supported formats are listed in tables.	Aug 14, 2018
V0.3	Pin List correction.	Aug 16, 2018
V0.4	Typo correction.	Aug 17, 2018
V0.5	Pin List correction again.	Aug 27, 2018
V0.6	Update HDR support features description.	Nov 5, 2018
V0.8	Add cascading stages information description.	Jul 11, 2019
V0.9	Update supported minimum operation temperature.	Jul 31, 2019
V1.0	Typo correction.	Sep 24, 2019
V1.1	Add typical soldering profile.	Feb 25, 2021
V2.0	Reformat the document.	Jun 24, 2021

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